

Here is the FloodSense-PK architecture and operational framework, restructured into a clean, hierarchical format for quick scanning and reference.

1. Institutional and Regulatory Framework

Disaster risk management in Pakistan operates under the National Disaster Management Act of 2010, utilizing a multi-tiered administrative structure:

- **National Disaster Management Commission (NDMC):** The primary national body for policymaking and plan approvals.
- **National Disaster Management Authority (NDMA):** The executive arm that implements and coordinates strategies at the national level.
- **Provincial/Regional Authorities:** PDMAs (Provinces), SDMAs (State), and GBDMAs (Gilgit-Baltistan) mirror the NDMA structure territorially.
- **Local Authorities:** DDMA's and Tehsil structures handle localized emergency rescue and rehabilitation.
- **Federal Flood Commission (FFC):** Oversees national flood protection and quantitative precipitation forecasting (QPF) under the Ten-Year National Flood Protection Plan-IV.
- **National Institute of Disaster Management (NIDM):** Drives research, database building, and personnel training in standards like e-MHVRA and INSaR.

The strategic planning evolution spans from NDMP-I (2012–2022) to the current NDMP 2025–2030, which integrates AI, GIS, and satellite monitoring into hazard mapping.

2. Regional Hydro-Meteorological Risks

Pakistan's diverse geography presents distinct, localized flood threats:

- **Gilgit-Baltistan:** Faces rapid glacial melt and glacial lake outburst floods (GLOFs) from 52 potentially dangerous lakes (e.g., Shishper Glacier).
- **Urban Centers (Karachi):** Stormwater management systems fail under heavy rainfall, requiring major drainage channel restoration.
- **Arid & Coastal (Balochistan):** Prone to severe flash flooding from torrential monsoons. Delay-action dams are highly vulnerable (e.g., the 2005 Shadikor Dam failure). Runoff from these catchments also drains east, threatening Sindh districts.
- **Kabul River Basin (Khyber Pakhtunkhwa):** A dense tributary network causes rapid, high-velocity flow accumulation. The 2010 and 2022 floods saw extreme peaks of up to 300,000 cusecs, devastating the Peshawar Valley.

3. Discharge Capacities and Hydrological Flow Thresholds

Riverine flooding in the Indus River Basin is managed using categorized discharge thresholds.

Key Barrage and Reservoir Thresholds

(Values in Lac Cusecs unless noted)

River	Structure	Capacity	Low	Medium	High	Very High	Exceptional	Historic

								Peak (Date)
Indus	Tarbel a Reser voir	15.00	2.50	3.75	5.00	6.50	8.00	604,0 00 (30-0 7-201 0)
Indus	Kalaba gh Barrag e	9.50	2.50	3.75	5.00	6.50	8.00	950,0 00 (14-07 -1942)
Indus	Tauns a Barrag e	10.00	2.50	3.75	5.00	6.50	8.00	959,99 1 (02-0 8-201 0)
Indus	Gudd u Barrag e	12.00	2.00	3.50	5.00	7.00	9.00	1,199,6 72 (15-08 -1976)
Indus	Sukkur Barrag e	9.00	2.00	3.50	5.00	7.00	9.00	1,161,0 00 (16-08 -1976)
Indus	Kotri Barrag e	8.75	2.00	3.00	4.50	6.50	8.00	981,00 0 (14-08 -1956)
Jhelu m	Mangl a Reser voir	12.02	0.75	1.10	1.50	2.25	3.00	952,17 0 (10-09 -1992)

Chenab	Marala Head works	11.00	1.00	1.50	2.00	4.00	6.00	1,100,000 (26-08-1957)
Chenab	Trimmu Head works	6.45	1.50	2.00	3.00	4.50	6.00	611,526 (Pre-breach 2010)

Critical Water Travel Times

Propagation intervals dictate available evacuation lead times.

River System	Upstream Point	Downstream Point	Travel Time
Ravi	Madhopur	Jassar	18 Hours
Chenab	Marala	Khanki	12 Hours
Chenab	Qadirabad	Trimmu	48 Hours
Jhelum	Rasul Barrage	Trimmu Headworks	72 Hours
Indus	Chashma	Taunsa	72 Hours
Indus	Taunsa	Mithankot + Guddu	48 + 24 Hours
Indus	Sukkur	Kotri	72 Hours
Sutlej	Islam	Panjinad	72 Hours

4. FloodSense-PK Technical Ingestion Pipeline

FloodSense-PK uses an automated, multi-stage pipeline to convert unstructured data into vector representations.

- **Extraction & Parsing Phase:**
 - **Unstructured PDFs:** OCR and layout-preserving parsers process policy documents to

- retain margins and headings.
- **Tabular PDFs/CSVs:** Structure-aware engines (like Camelot) isolate tabular grids to prevent cell merging.
- **Scientific Formulations:** Syntax-aware regex patterns isolate LaTeX math blocks to prevent splitting inline formulas.
- **Semantic Chunking Phase:**
 - **Hierarchical Chunking:** Policy files are split by markdown headings to keep sections unified.
 - **Row-Level Prepending:** Tabular data is divided row-by-row, with table headers and basin metadata dynamically prepended to each row.
 - **Command-Centric Chunking:** Emergency procedural manuals are grouped by "trigger-to-action" blocks so critical safety actions stay linked to their conditions.

5. Metadata Schema and Retrieval Architecture

Each pipeline chunk is tagged with structured metadata to enable precise relational filtering before semantic vectors are matched.

JSON

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{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "FloodSensePK_Metadata_Schema",
  "type": "object",
  "properties": {
    "document_id": {"type": "string"},
    "source_agency": {"type": "string"},
    "temporal_scope": {
      "type": "object",
      "properties": {
        "season": {"type": "string", "enum": ["Monsoon", "Winter", "Summer", "Non-Seasonal"]}
      }
    },
    "required": "season",
    "spatial_coverage": {
      "type": "object",
      "properties": {
        "province": {"type": "string"},
        "districts": {"type": "array", "items": {"type": "string"}}
      }
    }
  }
}
```

```

"required" "province"
"hydrological_parameters"
"type" "object"
"properties"
  "gauge_station" "type" "string"
  "discharge_cusecs" "type" "integer"

```

Advanced Indexing & Models

- **Hybrid Search:** Combines **Dense Semantic Indexing** (Bi-encoder transformer for conceptual queries) with **Sparse Lexical Indexing** (BM25 for precise numerical/technical term retrieval). Results are merged via Reciprocal Rank Fusion (RRF).
- **Multi-Vector Schemas:** Summaries of dense technical tables are indexed for semantic matching, but retrieve the full structural markdown table when triggered.
- **Satellite Segmentation:** The database ingests Sentinel-1 SAR models (like U-Net and Nested U-Net) to support real-time early warning capabilities during heavy monsoon cloud cover.

6. Operational Field Response and Protocols

Quantitative triggers physically mobilize ground teams across various jurisdictions.

- **PRCS Early Action Protocol (EAP):** Utilizes JBA Flood Foresight to release funds before peak flooding.
 - **Section 4 Trigger:** Trimmu Headworks $\geq 300,000$ cusecs (24-hour forecast) initiates warnings and evacuations.
 - **Section 5 Trigger:** Taunsa Barrage $> 550,000$ cusecs mobilizes warehoused logistical reserves.
- **Rescue 1122 & Municipal SOPs:** Under the Punjab Emergency Services Act, a declared emergency mobilizes 50% of the 7,500-person Disaster Response Force.
 - Equipment must be staged 15 feet from the water's edge.
 - Municipal bodies (WASAs, PHED, LG&CDD) must complete 100% desilting, activate dewatering pumps, and run 24/7 District Emergency Operations Centers (DEOCs).

7. Strategic Deployment Requirements

To ensure FloodSense-PK remains operationally reliable, the system enforces three core capabilities:

1. **Multi-Tiered Context Preservation:** The schema strictly maps hierarchical relationships so the LLM does not confuse federal guidelines (third-tier responders) with local DDMA directives (first-tier responders).

2. **Dynamic Temporal Boundaries:** All data carries strict temporal metadata (e.g., season, year) to prevent the retrieval engine from serving outdated compensation rates or obsolete contingency frameworks.
3. **Numerical Grounding:** The hybrid search index ensures the system explicitly differentiates between similar but distinct flow limits (e.g., 550,000 cusecs at Taunsa vs. 300,000 cusecs at Trimmu) without semantic distortion.